

Deep Learning FL25 Competition Report

Team Epoch-alyipse

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Abstract

Self-supervised learning (SSL) has emerged as a powerful tool for learning visual representations from large-scale unlabeled data, reducing reliance on costly human annotations while enabling strong transfer to downstream tasks. Recent advances have shown that carefully designed pre-training objectives, data augmentations, and optimization strategies can produce representations that perform well when evaluated under frozen-backbone protocols. In this work, we study SSL in a constrained but realistic setting: learning visual features from roughly 500k unlabeled images at a fixed resolution of 96×96 pixels, using a randomly initialized backbone with fewer than 100 million parameters. Our final model is a MoCo v2-style contrastive framework built on a ResNet-50 backbone, incorporating a momentum encoder, an enhanced MLP projection head, and an extended augmentation pipeline. Following standard practice, the pretrained encoder is evaluated without further adaptation using a linear probe on labeled downstream datasets. Our goal is to investigate how design choices in SSL training affect representation quality under strict evaluation and competition constraints.

1. Related Work

1.1. Contrastive Self-Supervised Learning

Contrastive self-supervised learning trains encoders to align representations of different augmented views of the same image while separating distinct images. Methods such as SimCLR and MoCo showed that strong augmentations and careful handling of negative samples enable effective large-scale representation learning [4, 5].

1.2. Non-Contrastive and Redundancy-Reduction Methods

Non-contrastive approaches eliminate explicit negatives, simplifying optimization. BYOL demonstrated that asymmetric networks with target encoders can learn meaning-

ful representations without collapse, while redundancy-reduction methods such as VICReg explicitly regularize feature statistics to stabilize training [1, 6].

1.3. Clustering and Distillation-Based Methods

Clustering- and distillation-based methods cast SSL as a prediction or assignment task. SwAV uses online clustering, while DINO applies self-distillation between student and teacher networks to learn semantically structured representations [2, 3].

1.4. Alternative SSL Paradigms

Alternative paradigms explore objectives beyond instance discrimination. Masked image modeling approaches such as MAE learn representations via patch reconstruction, while patch- and token-level methods introduce finer-grained supervisory signals [7].

Among the approaches discussed, this work experiments with MoCo and DINOv2 as representative contrastive and distillation-based self-supervised learning methods.

2. Methods

2.1. Self-Supervised Pretraining with MoCo

We adopt a self-supervised contrastive learning framework based on Momentum Contrast (MoCo) to learn visual representations from unlabeled data. Given an input image, two stochastically augmented views are generated and encoded using a query encoder and a momentum-updated key encoder. The model is trained to maximize agreement between representations of augmented views from the same image while distinguishing them from representations of other images stored in a large negative queue. This approach enables representation learning without reliance on labeled data.

2.2. Model Architecture

Our backbone network is a randomly initialized ResNet-50. Following the MoCo framework, we maintain two encoders with identical architectures: a query encoder updated

via backpropagation and a key encoder updated using an exponential moving average of the query encoder parameters. Outputs from the backbone are passed through a multi-layer perceptron (MLP) projection head. In the initial setup, the projection head outputs 128-dimensional embeddings, while in the final variant submitted, the embedding dimension is increased to 256 and the projection head incorporates Batch Normalization and ReLU activations.

2.3. Data Augmentation

For each image, two augmented views are sampled independently to form positive pairs. The core augmentations include random resized cropping, horizontal flipping, color jitter, and occasional grayscaling, which are standard in contrastive learning to encourage semantic invariance. In our final MoCo configuration, we further extend the pipeline with Gaussian blur and optional solarization, following MoCo v2+ practices, to promote stronger invariance and improve representation quality.

For our 96×96 images, we adjusted the augmentations from the original MoCo settings, which were designed for ImageNet-scale images. Specifically, we used a higher minimum crop scale, moderate color jitter, and smaller blur kernels to preserve semantic content at lower resolution. We also incorporated noise injection, which, while non-canonical, provided a modest performance boost by improving robustness to sensor noise and small perturbations.

2.4. Contrastive Objective

The model is trained using the InfoNCE contrastive loss. For each query embedding, the corresponding key embedding from the same image forms a positive pair, while all other embeddings in a fixed-size queue serve as negatives. We use a queue of approximately 65,000 negative examples, enabling efficient large-scale contrastive learning and improving representation discrimination.

2.5. Optimization and Training Details

Training is performed using stochastic gradient descent with a cosine learning rate schedule and mixed-precision arithmetic. The initial MoCo model is trained with a batch size of 512 for 50 epochs on the unlabeled image dataset. For the final MoCo variant, the batch size is reduced to 256 to improve training stability and accommodate memory constraints, while all other optimization settings and the contrastive objective remain unchanged.

3. Evaluation Protocol

3.1. Frozen-Encoder Protocol

The pretrained MoCo backbone is frozen during all downstream evaluation. No gradients are propagated to the encoder, and all tuning is restricted to the evaluation head

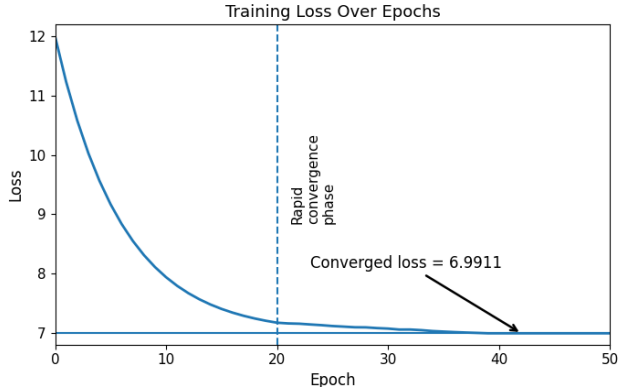


Figure 1. Training loss over 50 epochs for MoCo pretraining. Loss decreases rapidly during the first 20 epochs (dashed line), followed by a slow, noisy plateau as optimization approaches convergence. The final loss stabilizes around 6.99, with small epoch-to-epoch fluctuations characteristic of stochastic gradient descent. Notably, we found that training loss was not a reliable indicator of downstream performance: an alternative MoCo variant converged to a lower final loss (6.4471) yet failed to yield improved accuracy on any evaluation test set. This highlights that lower contrastive loss does not necessarily translate to better learned representations under frozen-backbone evaluation.

to ensure that performance reflects representation quality rather than task-specific adaptation.

3.2. Linear Probe Evaluation

We evaluate learned representations using a linear probe trained on features extracted from the frozen backbone. Specifically, we train a multinomial logistic regression classifier on the training split of the downstream dataset. Prior to probing, features are normalized using a standard scaler to stabilize optimization.

3.3. Hyperparameter Selection

Hyperparameters for the linear probe are selected using validation accuracy. We sweep the regularization strength C over a logarithmic grid from 0.01 to 100 and evaluate both unweighted and class-balanced loss formulations. The configuration achieving the highest validation accuracy is selected.

3.4. Test Evaluation

Using the selected hyperparameters, the linear probe is re-trained on the full training set and evaluated once on the test set. Test images are used strictly for inference, and no adaptation is performed during evaluation.

4. Results

We evaluate our self-supervised models using a frozen-backbone protocol on three held-out test sets. Performance

	Test Set 1	Test Set 2	Test Set 3
Public	0.1499	0.5167	0.2545
Private	0.1377	0.5243	0.2505

Figure 2. Public and private leaderboard accuracies for the final self-supervised model evaluated via a frozen-backbone linear probe across three test sets.

is measured using top-1 classification accuracy. On Test Set 1, the model achieves an accuracy of 13.77%, reflecting the difficulty of the task under limited supervision. Performance improves substantially on Test Set 2, where the model reaches 52.43% accuracy, indicating stronger transfer under this evaluation setting. On Test Set 3, the model achieves 25.05% accuracy, demonstrating moderate generalization performance across varying test conditions. These results highlight the sensitivity of downstream performance to evaluation setup and data characteristics, and underscore the importance of robust representation learning in self-supervised frameworks.

5. Discussion

5.1. Failed Attempts

Our initial version of this project explored an ambitious DINOv2-style knowledge distillation setup in which a frozen ViT-small DINOv2 teacher supervised a smaller, randomly initialized ViT student on the course’s 500k-image unlabeled corpus. The student was trained to regress to the teacher’s internal CLS representation with an MSE objective, using timm ViT backbones, mixed-precision training on an A100 GPU, and a Lightning-based data pipeline over the HuggingFace dataset. Although this setup was conceptually appealing and we were able to get it running end-to-end on a subset of the data, empirical profiling showed that each epoch required tens of hours: every training step performed both a teacher and student forward pass on top of relatively heavy data loading, so effective throughput was well below 1 iteration per second. Given the project’s time-sensitive deadline and limited GPU budget, this made full-dataset, multi-epoch pre-training effectively infeasible both in wall-clock time and in compute cost.

Before completely abandoning the idea of using a DINOv2 framework, we decided to pivot to a lighter weight backbone for the teacher and student models. The idea was that with strong augmentations and slow teacher momentum, we could make the DINOv2 architecture work with a ResNet-18 backbone.

We initially observed numerical instability when training DINOv2 with ResNet backbones under automatic mixed precision (AMP). Mixed-precision arithmetic exacerbates collapse by underflowing small logits and gradients in the

projection head, particularly in the presence of centering, temperature scaling, and softmax operations over high-dimensional output spaces, which together remove much of the representational signal. This interaction accelerates convergence toward trivial fixed points, especially for the low-variance feature representations produced by ResNet backbones, ultimately leading to vanishing gradients. While disabling AMP alleviated the numerical instability, it revealed the underlying architectural fragility of our setup rather than resolving the root cause of collapse.

Despite employing a larger batch size, correct multi-crop augmentation, training continued to consistently collapse, characterized by rapidly decreasing loss, vanishing gradient norms, and abnormally high teacher entropy. This behavior is indicative of a uniform collapse regime, in which teacher outputs converge to an uninformative, near-uniform distribution, eliminating the learning signal.

The failure of ResNet-18 in this setting stemmed from a mismatch between the properties of small convolutional networks and the requirements of self-distillation without labels. DINO depends on meaningful differences between multiple views of the same image early in training in order to produce a useful learning signal. In contrast, ResNet-18 is designed to make representations similar across views from the start. Local convolution, early pooling, weight sharing, and Batch Normalization all encourage this behavior, reducing variation in the learned features before semantic structure has time to emerge. As a result, the feature space becomes overly compressed, and the centering and temperature scaling used in DINO quickly remove what little structure remains. As a result, the teacher entropy exploded epoch-over-epoch, leading to vanishing gradients.

Although models such as ResNet-50 have been used in earlier DINO work, their stability arises from higher representational capacity and feature rank. Deeper CNNs preserve more diversity across views and are less susceptible to over-normalization effects induced by centering and Batch Normalization. In contrast, ResNet-18 was too small, lying near a capacity threshold, where aggressive invariance and limited expressivity combined to make uniform collapse highly likely.

In contrast, ViTs align naturally with the objectives of DINO. ViTs preserve patch-level diversity through tokenized representations, delay the emergence of invariance due to the absence of pooling, and maintain high-rank embedding spaces through self-attention. As a result, centering and temperature sharpening reduce bias without destroying meaningful structure, allowing teacher outputs to remain informative and gradients to persist throughout training. Although ResNet backbones offer faster iteration speed and lower memory usage, these advantages were outweighed by their fragility in self-distillation settings. After many iterations of parameter tuning and failed training due to

collapse, we decided to move away from pursuing the knowledge distillation track, as training with a light-weight ResNet backbone was still time-intensive relative to other self-supervised options.

As a result, we concluded that the DINOv2 distillation route was the wrong fit for our constraints and pivoted to a more lightweight MoCo v2 baseline that could be trained to convergence on the full dataset within the available time and resources.

6. Future Work

Given additional time, we would prioritize practical improvements to MoCo training, beginning with more thorough learning rate optimization. In particular, we would conduct a systematic sweep over initial learning rates (e.g., 0.03–0.3) and learning rate schedules, including linear warmup over the first 5–10 epochs followed by cosine decay. This is especially important in contrastive self-supervised learning, where training stability and representation quality are highly sensitive to early optimization dynamics, as also suggested in the course materials.

In our final training runs, we observed that even after 40+ epochs the contrastive loss continued to decrease, albeit slowly. This behavior suggests that our learning rate schedule may have been overly aggressive: starting from a relatively small initial learning rate and decaying it rapidly resulted in an effective learning rate that became too small to support meaningful progress in later epochs.

To sanity-check this hypothesis, we replaced the larger ResNet-50 backbone with a smaller and faster ResNet-18 backbone and trained it using a higher initial learning rate. Despite its lower capacity, the ResNet-18 model achieved slightly better performance than the ResNet-50 model trained with a small learning rate. This result indicates that optimization dynamics, rather than model capacity, were a limiting factor in our experiments.

These findings suggest that the underperformance of the ResNet-50 MoCo model was likely due to suboptimal learning rate choices rather than architectural limitations. With a larger backbone, a higher initial learning rate combined with warmup and a less aggressive decay schedule would likely yield stronger representations and improved downstream performance.

Second, we would test larger queue sizes (100k+) and fine-tune momentum ($m=0.995$ – 0.999) for more robust negative sampling. Third, multi-crop augmentations (SwAV-style) would increase positive pair diversity with minimal compute. We would also use additional unlabeled competition data for extended pretraining if needed, and perform per-test-set error analysis to address the 13.77%–52.43% variance.

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